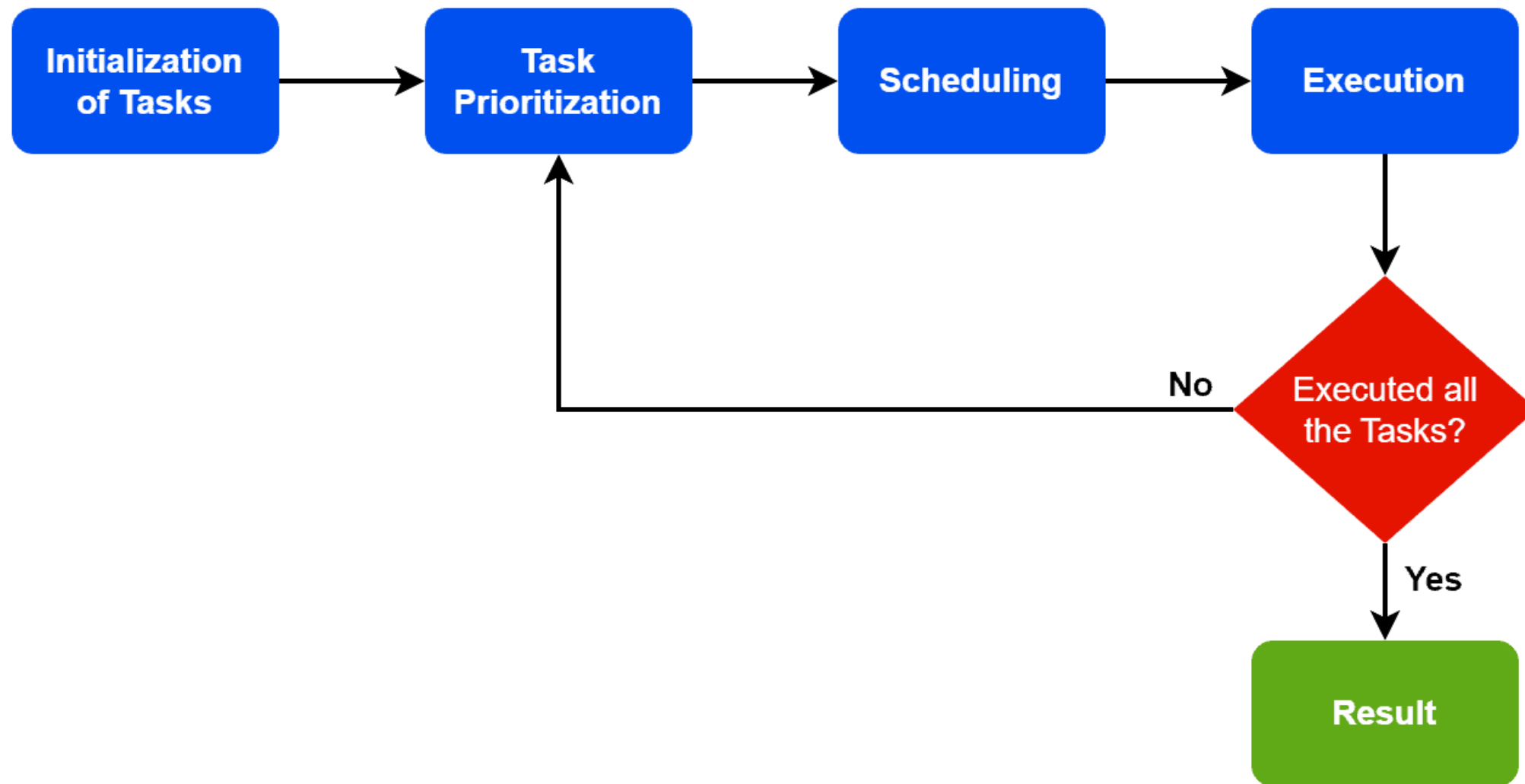


Earliest Dead Line First Scheduling

- Earliest deadline first (EDF) comes under the category of the dynamic scheduling algorithm.
- We can use it in real-time operating systems for scheduling tasks as well as processes with specific deadlines.
- **It's a priority-based algorithm where we assign the highest priority to the tasks with the earliest deadline.**
- Additionally, the CPU executes the scheduled tasks based on priority.
- The algorithm assigns a priority to each task based on its deadline and chooses the task with the earliest deadline for execution.
- Additionally, if multiple tasks have the same deadline, the CPU executes the task with the highest priority.
- **EDF is an optimal algorithm for scheduling a set of periodic tasks with known deadlines** and execution times as long as the total utilization of the tasks doesn't exceed unity.
- Hence, the total execution time of all tasks should be less than the total available time



- **The earliest deadline first scheduling algorithm consists of four steps: initialization, task prioritization, task scheduling, and task execution.**
- The first step is to initialize the available tasks. Additionally, along with initialization, we assign each task a deadline based on completion requirements.
- The next step is to assign priority to each task. The system sorts the tasks in order of their deadlines, with the task having the earliest deadline assigned the highest priority.
- Furthermore, the system selects the task with the earliest deadline for execution. If multiple tasks have the same deadline, we select the task with the highest priority.
- Finally, the CPU executes the selected task until completed or the deadline is reached.
- If the task is completed before its deadline, the system returns to the task prioritization step to select the next task for execution.
- If the deadline is reached, the task is considered to be missed.

- Now before the algorithm terminates, we need to check two conditions.
- First, we need to check whether all the tasks are executed.
- Additionally, we check if all deadlines have been missed. If one of these conditions is met, we need to go back to the task prioritization step and repeat the steps until the algorithm terminates.
- EDF is a dynamic scheduling algorithm where the task priority can change over time as deadlines approach.
- Hence, the algorithm continually updates the priority of tasks based on their deadlines and selects the task with the earliest deadline for execution.

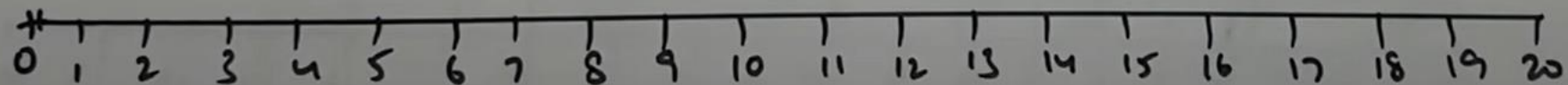
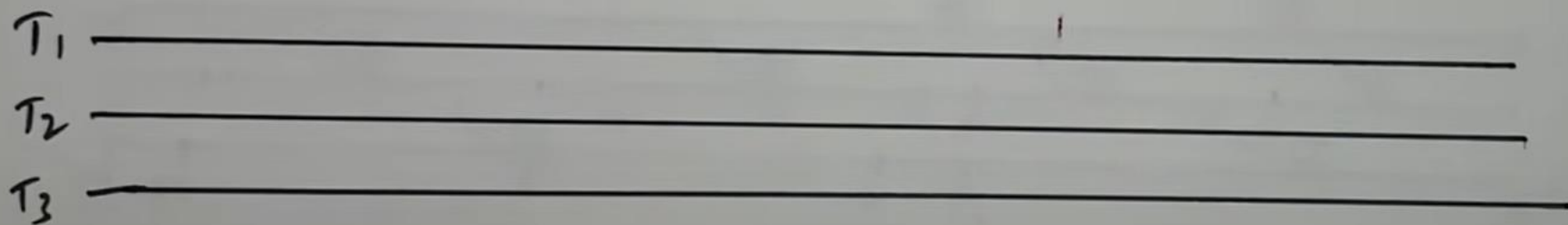
Earliest Deadline First (EDF)

- **Schedulability check (off-line)**
 - A set of n tasks is schedulable on a uniprocessor by the EDF algorithm if the processor utilization.

$$\sum_{i=1}^n c_i/p_i \leq 1$$

- This condition is both necessary and sufficient.
 - Least Laxity First (LLF) algorithm has the same schedulability check.

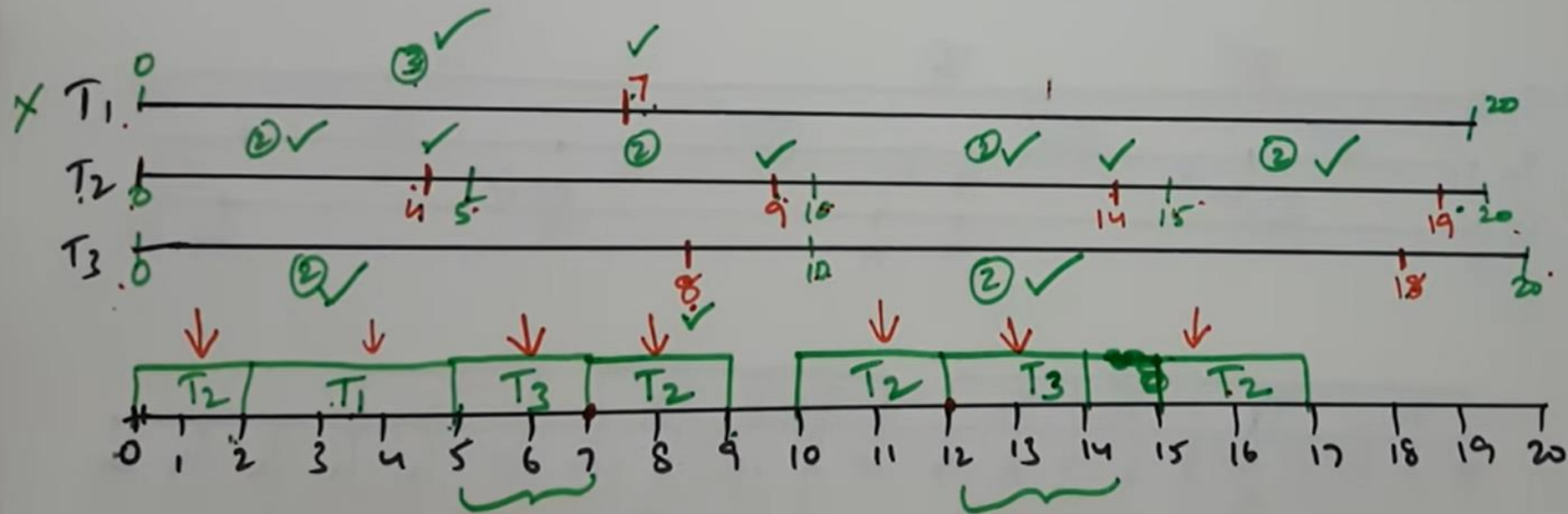
	<u>Capacity</u>	<u>Deadline</u>	<u>Period</u>
T_1	3	7	20
T_2	2	4	5
T_3	2	8	10



Earliest Deadline First scheduling

	Capacity	Deadline	Period
T_1	3	7	20
T_2	2	4	5
T_3	2	8	10

$$\frac{L.C.M(20, 5, 10)}{20 \text{ units}}$$



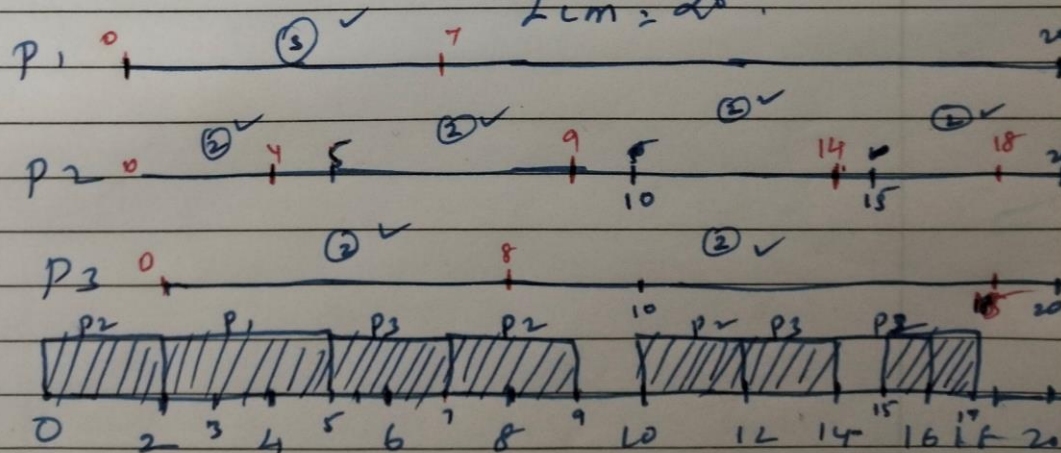
Earliest Deadline First Scheduling.

Example:

Process	Capacity	Deadline	Period.
P ₁	3	7	20
P ₂	2	4	5
P ₃	2	8	10.

$Lcm(20, 5, 10)$

$Lcm = 20$



- CPU utilization
 - Total time availability is 20
 - $P1 = 1 * 3 = 3$
 - $P2 = 4 * 2 = 8$
 - $P3 = 2 * 2 = 4$
 - Usability is $3 + 8 + 4 = 15$
 - CPU utilization is $15 / 20 = 75\%$

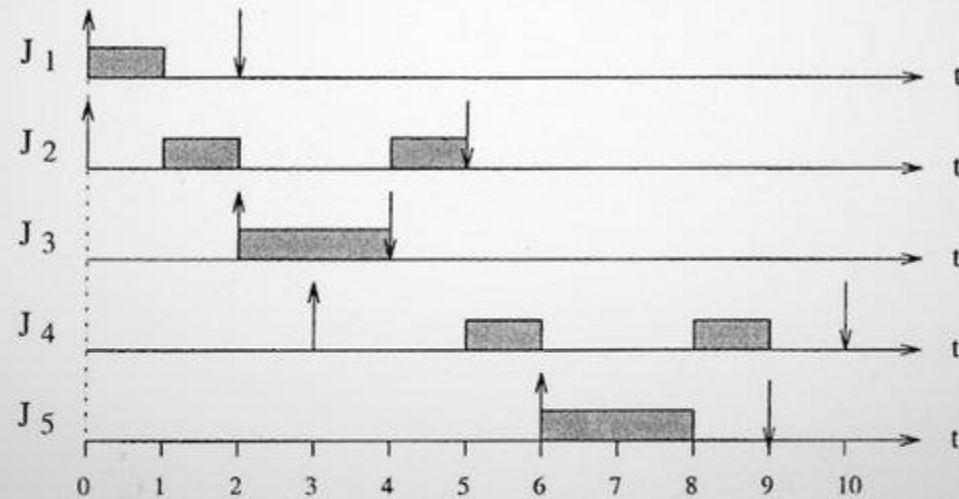
Example 2

Deadline is not mentioned hence
Period=Deadline

Process	Execution Time	Period
P ₁	1	3
P ₂	1	4
P ₃	2	8

Earliest Deadline First (EDF) – Example

	J_1	J_2	J_3	J_4	J_5
a_i	0	0	2	3	6
C_i	1	2	2	2	2
d_i	2	5	4	10	9



Example of EDF schedule.

Question 4 (6 points): For the following real-time processes, draw the Gantt chart, calculate the CPU utilization. Then, decide whether the two processes would miss their deadlines or not using:

- a. Rate Monotonic (RM) algorithm
- b. Earliest Deadline First (EDF) algorithm

P1: Period = 75, Processing Time = 35

P2: Period = 40, Processing Time = 20